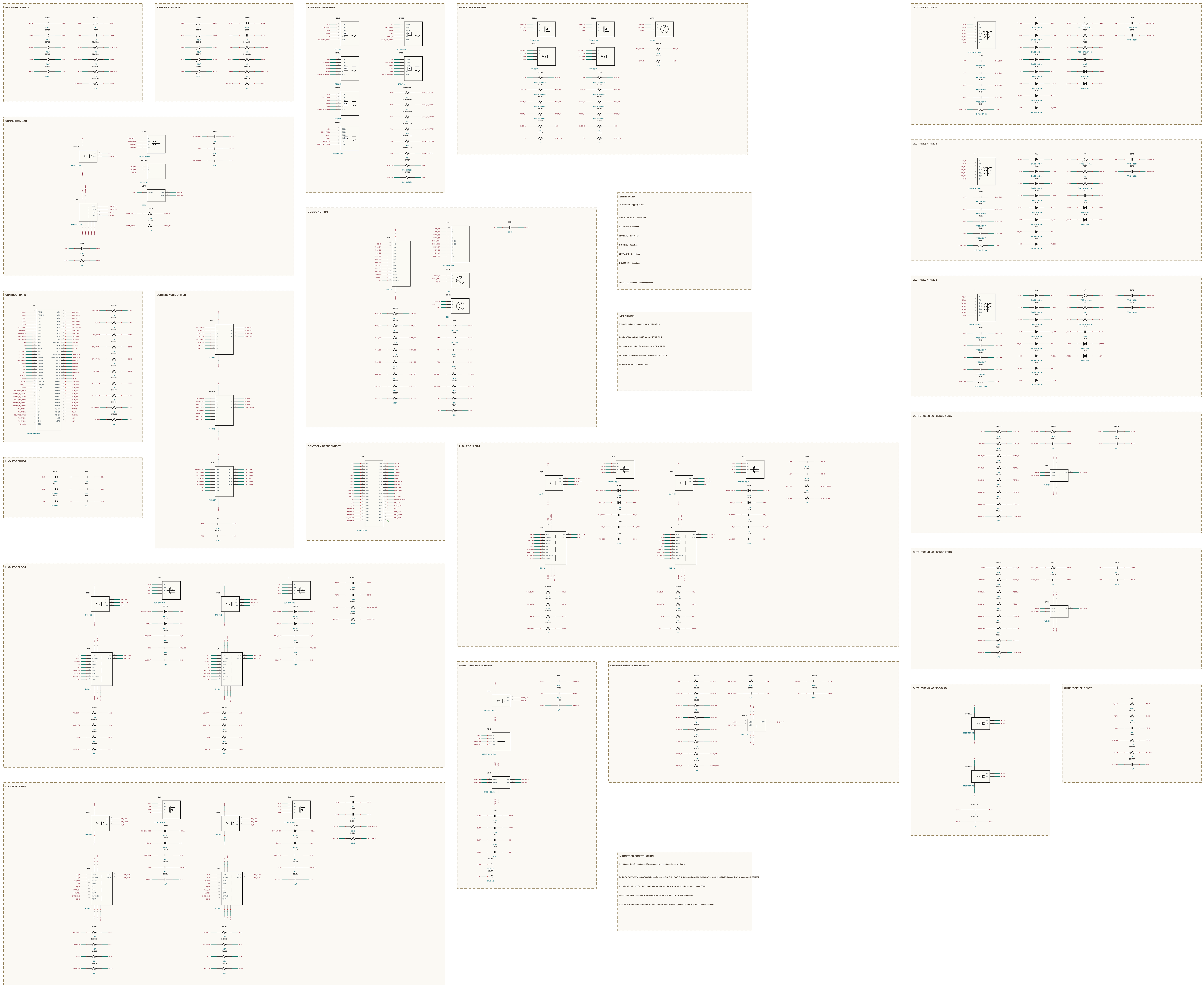


SHEET INDEX

40 kW DC-DC (upper) - 2 of 3

OUTPUT-SENSING - 6 sections

BANKS-SP - 4 sections

LLC-LEGS - 4 sections

CONTROL - 3 sections

LLC-TANKS - 3 sections

COMMS-HMI - 2 sections

rev D.4 - 22 sections - 322 components

NET NAMING

internal junctions are named for what they join

U<ref>_<PIN> node at that IC pin e.g. UIVOA_VINP

R<stem>_M midpoint of a series pair e.g. RBALTA_M

R<stem>_<nm> tap between R<stem>n/m e.g. RV1D_01

all others are explicit design nets

MAGNETICS CONSTRUCTION

identity per docs/magnetics.md (turns, gap, litz, acceptance lines live there)

D3 T1-T3: 2x E70/33/32 sets (B66372B2000 former), 6:6:6, Bpk 176mT @525V-bank sim, pri litz 3486x0.071 + sec foil 0.127x28, Lm 63uH +/-7% gap-ground, BONDED

D2 L1T-L3T: 2x E70/33/32, N=5, bins 5.85/6.0/6.15/6.3uH, litz 8149x0.05, distributed gap, bonded (E65)

total Lr = D2 bin + measured xfmr leakage (~0.2uH) + 0.1uH loop; Cr at TANK sections

T_XFMR NTC loop runs through 6 NC 130C cutouts, one per D3/D2 (open loop = OT trip, E65 bond-loss cover)

40 kW DCDC board 2of2 - 3-phase LLC (1x hot cells)

DC-Modules 40 kW - board DC-DC (upper), sheet 2 of 3

2026-09-06 · rev D.4